

# Basis Sets & Integrals

Pierre-François Loos

Laboratoire de Chimie et Physique Quantiques (UMR 5626), Université de Toulouse, CNRS, UPS, Toulouse, France

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# Expanding the molecular orbitals

## Linear combination of basis functions

$$\psi_p(\mathbf{r}) = \sum_{\mu=1}^K C_{\mu p} \phi_{\mu}(\mathbf{r})$$

- The basis functions  $\{\phi_{\mu}\}$  are **fixed**; the coefficients  $C_{\mu p}$  are determined **variationally**
- Increasing  $K$  improves the flexibility and, ideally, approaches the **complete basis set limit (CBS)**
- Basis sets contain the **only empirical parameters** of an *ab initio* calculation

## A useful basis set should be...

**Accurate** Flexible enough to describe the relevant occupied and virtual orbitals

**Compact** Able to reach the target accuracy with as few functions as possible

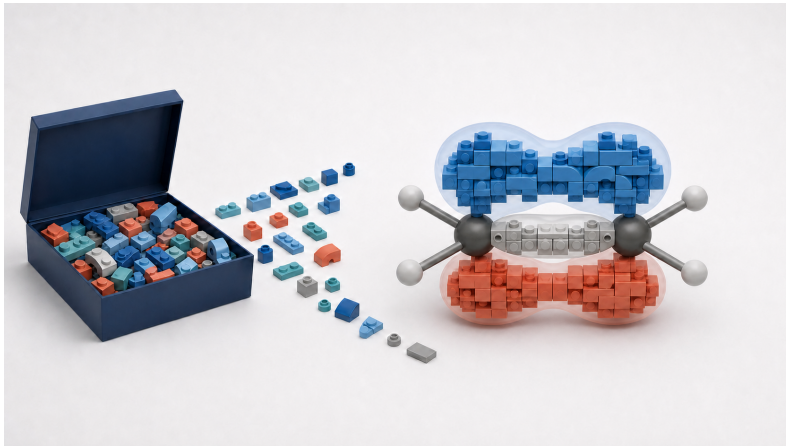
**Stable** Free from severe linear dependencies

**Practical** Compatible with evaluation of multicenter one- and two-electron integrals

⇒ **Basis set design is a compromise between accuracy and cost**

# Basis functions are like LEGO bricks!

**Key idea:** Each basis function is a building block



# Common basis functions

## Gaussian-type orbitals (GTOs)

$$\phi(\mathbf{r}) = \exp(-\alpha r^2)$$

- Usually atom-centered
- Standard choice for molecular calculations
- Efficient evaluation of multicenter integrals
- Software: Orca, Q-Chem, Molcas, Molpro, PySCF, Psi4, CFOUR, MRCC, GAMESS, etc

## Plane waves (PWs)

$$\phi(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r})$$

- Delocalized and orthonormal
- Naturally adapted to periodic systems
- Systematically converged with an energy cutoff
- Software: Quantum ESPRESSO, VASP, ABINIT, CASTEP, etc

## Slater-type orbitals (STOs)

$$\phi(\mathbf{r}) = \exp(-\zeta r)$$

- Usually atom-centered
- Natural nuclear cusp and exponential tail
- Software: ADF

## Numerical atomic orbitals (NAOs)

- Radial functions tabulated numerically
- Compact and flexible atom-centered basis
- Software: FHI-aims, SIESTA, etc

## Definition

$$\phi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} \exp(i\mathbf{k} \cdot \mathbf{r}) \quad \text{with} \quad |\mathbf{k}| \leq k_{\text{cut}} \quad \text{and} \quad E_{\text{cut}} = \frac{1}{2}k_{\text{cut}}^2$$

## Reciprocal-lattice vector

$$\mathbf{k} = k_1 \mathbf{b}_1 + k_2 \mathbf{b}_2 + k_3 \mathbf{b}_3$$

## Unit cell

$$\underbrace{\Omega}_{\text{volume}} = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3) \quad \text{with} \quad \mathbf{A} = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$$

## Vectors of the reciprocal lattice

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\Omega}$$

$$\mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\Omega}$$

$$\mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\Omega}$$

# Integrals in the plane wave basis

## Overlap integrals

$$\langle \mathbf{k}_1 | \mathbf{k}_2 \rangle = \int \phi_{\mathbf{k}_1}^*(\mathbf{r}) \phi_{\mathbf{k}_2}(\mathbf{r}) d\mathbf{r} = \frac{1}{\Omega} \int e^{i(\mathbf{k}_2 - \mathbf{k}_1) \cdot \mathbf{r}} d\mathbf{r} = \delta_{\mathbf{k}_1 \mathbf{k}_2} \Rightarrow \text{Orthogonal basis!}$$

## Kinetic integrals

$$\langle \mathbf{k}_1 | -\frac{\nabla^2}{2} | \mathbf{k}_2 \rangle = \int \phi_{\mathbf{k}_1}^*(\mathbf{r}) \left( -\frac{\nabla^2}{2} \right) \phi_{\mathbf{k}_2}(\mathbf{r}) d\mathbf{r} = \int \phi_{\mathbf{k}_1}^*(\mathbf{r}) \frac{k_2^2}{2} \phi_{\mathbf{k}_2}(\mathbf{r}) d\mathbf{r} = \frac{k_2^2}{2} \delta_{\mathbf{k}_1 \mathbf{k}_2} \Rightarrow \text{Diagonal!}$$

## Electron repulsion integrals (ERIs)

$$\begin{aligned} \langle \mathbf{k}_1 \mathbf{k}_2 | \mathbf{k}_3 \mathbf{k}_4 \rangle &= \iint \phi_{\mathbf{k}_1}^*(\mathbf{r}_1) \phi_{\mathbf{k}_2}^*(\mathbf{r}_2) r_{12}^{-1} \phi_{\mathbf{k}_3}(\mathbf{r}_1) \phi_{\mathbf{k}_4}(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2 \\ &= \frac{1}{\Omega^2} \iint \frac{e^{i(\mathbf{k}_3 - \mathbf{k}_1) \cdot \mathbf{r}_1} e^{i(\mathbf{k}_4 - \mathbf{k}_2) \cdot \mathbf{r}_2}}{r_{12}} d\mathbf{r}_1 d\mathbf{r}_2 \end{aligned}$$

# Resolution of the Coulomb operator

## Coulomb operator

$$\frac{1}{r_{12}} = \frac{1}{\Omega} \sum_{\mathbf{k} \neq 0} \frac{4\pi}{k^2} e^{i\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)}$$

Dealing with the points where  $\mathbf{k} = \mathbf{0}$  is a bit tricky but doable...

## Electron repulsion integrals

$$\begin{aligned} \langle \mathbf{k}_1 \mathbf{k}_2 | \mathbf{k}_3 \mathbf{k}_4 \rangle &= \frac{1}{\Omega^3} \sum_{\mathbf{k} \neq 0} \frac{4\pi}{k^2} \iint e^{i(\mathbf{k}_3 - \mathbf{k}_1 - \mathbf{k}) \cdot \mathbf{r}_1} e^{i(\mathbf{k}_4 - \mathbf{k}_2 + \mathbf{k}) \cdot \mathbf{r}_2} d\mathbf{r}_1 d\mathbf{r}_2 \\ &= \frac{1}{\Omega} \sum_{\mathbf{k} \neq 0} \frac{4\pi}{k^2} \delta_{\mathbf{k}_3 - \mathbf{k}_1 - \mathbf{k}, 0} \delta_{\mathbf{k}_4 - \mathbf{k}_2 + \mathbf{k}, 0} \\ &= \frac{4\pi}{\Omega |\mathbf{k}_1 - \mathbf{k}_3|^2} \delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4} \Rightarrow \text{Momentum conservation!} \end{aligned}$$

# Plane waves: strengths and limitations

**Key idea:** Assemblies of atoms are slight distortions of free electrons

## Advantages

- ✓ Orthonormal and independent of atomic positions
- ✓ Naturally compatible with periodic boundary conditions
- ✓ Systematic convergence through a single parameter  $E_{\text{cut}}$
- ✓ Kinetic and Coulomb operators are diagonal in reciprocal space
- ✓ Efficient evaluation using fast Fourier transforms
- ✓ No basis-set superposition error or atom-centered Pulay forces

## Limitations

- ✗ Large basis sets:

$$N_{\text{PW}} \propto \Omega E_{\text{cut}}^{3/2}$$

- ✗ Core orbitals require prohibitively large cutoffs  $\Rightarrow$  pseudopotentials
- ✗ Computational cost increases with supercell size
- ✗ Delocalized basis functions provide little real-space sparsity
- ✗ Large virtual spaces make correlated methods expensive

$\Rightarrow$  Particularly effective for periodic valence-electron calculations



## Slater-type orbitals (STOs)

$$\phi_{\zeta n \ell m}^{\text{STO}}(\mathbf{r}) \propto r^{n-1} e^{-\zeta r} Y_{\ell m}(\hat{\mathbf{r}})$$

- ✓ It can model the electron-nucleus cusp!

$$\phi_{1s}^{\text{STO}}(r) \propto e^{-\zeta r} \Rightarrow \left. \frac{d\phi_{1s}^{\text{STO}}(r)}{dr} \right|_{r=0} = -\zeta$$

- ✓ Correct long-range behavior for IP!

$$\phi_{\text{HOMO}}^{\text{STO}}(r) \propto e^{-\sqrt{-2\epsilon_{\text{HOMO}}}r} \Rightarrow \text{IP} = -\epsilon_{\text{HOMO}}$$

- ✗ The multicenter integrals are **hard** to compute!

## Gaussian-type orbitals (GTOs)

$$\phi_{\alpha n \ell m}^{\text{GTO}}(\mathbf{r}) \propto r^{2(n-1)-\ell} e^{-\alpha r^2} Y_{\ell m}(\hat{\mathbf{r}})$$

- ✗ They are flat at the electron-nucleus coalescence point!

$$\phi_{1s}^{\text{GTO}}(r) \propto e^{-\alpha r^2} \Rightarrow \left. \frac{d\phi_{1s}^{\text{GTO}}(r)}{dr} \right|_{r=0} = 0$$

- ✗ They decay too rapidly at large  $r$ !

- ✓ The multicenter integrals are **easy** to compute!

⇒ GTOs usually win but we use more!

## Cartesian Gaussians

$$\text{Primitive GTO} = |\mathbf{a}\rangle = (x - A_x)^{a_x} (y - A_y)^{a_y} (z - A_z)^{a_z} e^{-\alpha |\mathbf{r} - \mathbf{A}|^2}$$

$$\text{Contracted GTO} = |\mathbf{a}\rangle \equiv \phi_{\mathbf{a}}^{\mathbf{A}}(\mathbf{r}) = \sum_k^K D_k |\mathbf{a}\rangle_k$$

- **Exponent:**  $\alpha$
  - **Center:**  $\mathbf{A} = (A_x, A_y, A_z)$
  - **Angular momentum:**  $\mathbf{a} = (a_x, a_y, a_z)$
  - **Total angular momentum:**  $a = a_x + a_y + a_z$
- 
- $p$  shell:  $p_x = (1, 0, 0)$     $p_y = (0, 1, 0)$     $p_z = (0, 0, 1)$
  - $d$  shell:  $(2, 0, 0), (0, 2, 0), (0, 0, 2), (1, 1, 0), (1, 0, 1), (0, 1, 1) \Rightarrow 6 \text{ } d \text{ cartesian functions!}$
  - $f$  shell:  $(3, 0, 0), (0, 3, 0), (0, 0, 3), (2, 1, 0), (2, 0, 1), (1, 2, 0), (0, 2, 1), (0, 1, 2), (1, 0, 2), (1, 1, 1) \Rightarrow 10 \text{ } f \text{ cartesian functions!}$

# Cartesian vs Spherical Gaussians

## Cartesian Gaussians

For a shell of angular momentum  $\ell$

$$N_{\ell}^{\text{cart}} = \frac{(\ell + 1)(\ell + 2)}{2}$$

| $\ell$                   | $s$ | $p$ | $d$ | $f$ |
|--------------------------|-----|-----|-----|-----|
| $N_{\ell}^{\text{cart}}$ | 1   | 3   | 6   | 10  |

For instance, a Cartesian  $d$  shell contains

$$\{x^2, y^2, z^2, xy, xz, yz\}$$

## Spherical Gaussians

For a pure angular momentum  $\ell$

$$N_{\ell}^{\text{sph}} = 2\ell + 1$$

| $\ell$                  | $s$ | $p$ | $d$ | $f$ |
|-------------------------|-----|-----|-----|-----|
| $N_{\ell}^{\text{sph}}$ | 1   | 3   | 5   | 7   |

The extra Cartesian functions correspond to  
lower-angular-momentum components:

$$6d = 5d + 1s \quad 10f = 7f + 3p$$

$x^2 + y^2 + z^2 = r^2 \Rightarrow$  the Cartesian  $d$  shell contains an  $s$ -type component

$\Rightarrow$  Most quantum-chemistry calculations use spherical (pure) functions

# The contraction problem

## Primitive vs Contracted

- Same center  $A$
- Same angular momentum  $a$
- Different exponent  $\alpha_k$
- Contraction coefficient  $D_k$  and degree  $K$

$$\underbrace{\langle ab|cd \rangle}_{\text{contracted ERI}} = \sum_i^{K_A} \sum_j^{K_B} \sum_k^{K_C} \sum_l^{K_D} D_i D_j D_k D_l \underbrace{[a_i b_j | c_k d_l]}_{\text{primitive ERI}}$$

One contracted ERI required  $K_A \times K_B \times K_C \times K_D$  primitive ERIs!

$$|a\rangle = \sum_k^K D_k |a_k\rangle$$

<https://www.basissetexchange.org>

|   |              |               |
|---|--------------|---------------|
| C | 0            |               |
| S | 10           | 1.00          |
|   | 8.236000D+03 | 5.310000D-04  |
|   | 1.235000D+03 | 4.108000D-03  |
|   | 2.808000D+02 | 2.108700D-02  |
|   | 7.927000D+01 | 8.185300D-02  |
|   | 2.559000D+01 | 2.348170D-01  |
|   | 8.997000D+00 | 4.344010D-01  |
|   | 3.319000D+00 | 3.461290D-01  |
|   | 9.059000D-01 | 3.937800D-02  |
|   | 3.643000D-01 | -8.983000D-03 |
|   | 1.285000D-01 | 2.385000D-03  |

## Dunning's cc-pVTZ basis for the carbon atom

$$\langle 1s1s|1s1s \rangle = \sum_i^{10} \sum_j^{10} \sum_k^{10} \sum_l^{10} D_i D_j D_k D_l [s_i^{\alpha_i} s_j^{\alpha_j} | s_k^{\alpha_k} s_l^{\alpha_l}]$$

The  $\langle 1s1s|1s1s \rangle$  integral requires  $10^4$  s-type integrals!

# Choose the basis for the physics you need

## Basis set choice is tricky

- **Valence flexibility:** use double-, triple-, or higher- $\zeta$  descriptions to give each valence orbital several radial shapes
- **Polarization functions:** add **higher angular momenta** to describe anisotropic density, bonding, and angular correlation
- **Diffuse functions:** add **small exponents** to describe long-range density (anions, Rydberg states, noncovalent interactions, multipoles, (hyper)polarizabilities, etc)
- **Tight functions:** add **large exponents** to describe core region (core correlation, nuclear magnetic shielding, spin-spin coupling constants, etc)

⇒ The appropriate basis depends on both the electronic structure method and the property of interest!

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2ZaP  
 2ZaPa-NR  
 2ZaPa-NR-CV  
 3-2IG  
 3ZaP  
 3ZaPa-NR  
 3ZaPa-NR-CV  
 4-3IG  
 4ZaP  
 4ZaPa-NR  
 4ZaPa-NR-CV  
 5-2IG  
 5ZaP  
 5ZaPa-NR  
 5ZaPa-NR-CV  
 6-2IG  
 6-3I+G  
 6-3I+G\*  
 6-3I+G\*\*  
 6-3I+G  
 6-3I+G\*  
 6-3I+G\*\*

search basis sets...

Total found: 776 basis sets

Select All
Reset Selection

|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
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|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Al | Si | P  | S  | Cl | Ar |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | K  | Ca | Sc | Ti | V  | Cr | Mn | Fe | Co | Ni | Cu | Zn | Ga | Ge | As | Se | Br | Kr |    |    |    |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Rb | Sr | Y  | Zr | Nb | Mo | Tc | Ru | Rh | Pd | Ag | Cd | In | Sn | Sb | Te | I  | Xe |    |    |    |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Cs | Ba |    | Hf | Ta | W  | Re | Os | Ir | Pt | Au | Hg | Tl | Pb | Bi | Po | At | Rn |    |    |    |
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|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | La | Ce | Pr | Nd | Pm | Sm | Eu | Gd | Th | U  | Np | Pu | Am | Cm | Bk | Cf | Es | Fm | Md | No | Lr |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

In the selected basis set

References for selected basis

**Description:**

**Latest Version:**

**Last Update:**

**Role:**

**Family:**

**Function Types:**

Basis Set Notes
Family Notes

Download basis set

**Citation**

When publishing results obtained from use of the Basis Set Exchange software, please cite:

- A New Basis Set Exchange: An Open, Up-to-date Resource for the Molecular Sciences Community. Benjamin P. Pritchard, Doaa Altarawy, Brett Didier, Tara D. Gibson, Theresa L. Windus. *J. Chem. Inf. Model.* **2019**, 59(7), 4814–4820, doi:10.1021/acs.jcim.9b00725.

For citing the previous EMSL/PNNL Basis Set Exchange, please cite the following references:

- The Role of Databases in Support of Computational Chemistry Calculations Feller, D., *J Comp. Chem.* **1996**, 17(13), 1571–1586.
- Basis Set Exchange: A Community Database for Computational Sciences Schuchardt, K.L., Didier, B.T., Eshelberger, T., Sun, L., Gurunoorith, V., Chase, J., Li, J., and Windus, T.L. *J. Chem. Inf. Model.* **2007**, 47(3), 1045–1052, doi:10.1021/ci600510j.

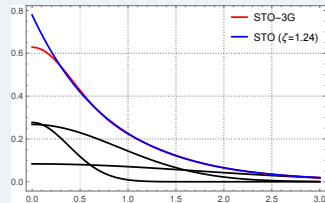
Basis Set Exchange, BSE Library v0.12  
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## Minimal basis set STO- $n$ G

- One contracted basis function for each occupied orbital in the minimal atomic description
- Each contracted GTO is made of  $n$  primitive GTOs
- The contraction coefficients and exponents are chosen so that the contraction approximates a STO
- STO- $n$ G basis sets are therefore **minimal**, **single- $\zeta$**  basis sets

## STO-3G for the H atom

```
!-----  
! Basis Set Exchange  
! Version 0.12  
! https://www.basissetexchange.org  
!-----  
! Basis set: STO-3G  
! Description: STO-3G Minimal Basis (3 functions/AO)  
!           Role: orbital  
!           Version: 1 (Data from Gaussian09)  
!-----  
  
H      0  
S      3      1.00  
        0.3425250914D+01      0.1543289673D+00  
        0.6239137298D+00      0.5353281423D+00  
        0.1688554040D+00      0.4446345422D+00  
****
```



# Pople basis sets

- **6-31+G\***

- **6** six primitive GTOs for each core orbital
- **31** two sets of contracted GTOs (double- $\zeta$ ) for each valence orbital with contraction degrees of 3 and 1
- **+** one set of diffuse functions on non-H atoms
- **\*** one set of polarization functions on non-H atoms [ $*$   $\equiv$  (d)]

- **6-311++G\*\***

- **311** three sets of contracted GTOs with different exponents (triple- $\zeta$ )
- **++** same as **+** but with diffuse functions on H atoms
- **\*\*** same as **\*** but with polarization functions on H atoms

- **6-311G(2df,2pd)**

- **2df** two sets of  $d$ 's and one set of  $f$ 's for non-H atoms
- **2pd** two sets of  $p$ 's and one set of  $d$ 's for the H atoms



## Correlation-consistent (cc) hierarchy

cc-pVXZ

$X = \text{D, T, Q, 5, 6, } \dots$

- Designed primarily for correlated wave function methods
- Polarization functions are added in balanced groups
- Increasing  $X$  systematically improves the valence space

## Two-point extrapolation model

$$E_c(X) \approx E_c^{\text{CBS}} + AX^{-3}$$

## Diffuse density

- aug-cc-pVXZ
- d-aug-cc-pVXZ
- t-aug-cc-pVXZ

## Core correlation

- cc-pCVXZ: Optimized to recover both core-core and core-valence correlation
- cc-pwCVXZ: Uses a weighted optimization that emphasizes core-valence correlation

## Correlation consistent basis for 2nd-row atoms

cc-pVDZ: 3s2p1d

|   |   |   |
|---|---|---|
| s | s | s |
| p | p |   |
| d |   |   |

aug-cc-pVDZ: 4s3p2d

|   |   |   |   |
|---|---|---|---|
| s | s | s | s |
| p | p | p |   |
| d | d |   |   |

d-aug-cc-pVDZ: 5s4p3d

|   |   |   |   |   |
|---|---|---|---|---|
| s | s | s | s | s |
| p | p | p | p |   |
| d | d | d |   |   |

cc-pVTZ: 4s3p2d1f

|   |   |   |   |
|---|---|---|---|
| s | s | s | s |
| p | p | p |   |
| d | d |   |   |
| f |   |   |   |

aug-cc-pVTZ: 5s4p3d2f

|   |   |   |   |   |
|---|---|---|---|---|
| s | s | s | s | s |
| p | p | p | p |   |
| d | d | d |   |   |
| f | f |   |   |   |

d-aug-cc-pVTZ: 6s5p4d3f

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| s | s | s | s | s | s |
| p | p | p | p | p |   |
| d | d | d | d |   |   |
| f | f | f |   |   |   |

cc-pVQZ: 5s4p3d2f1g

|   |   |   |   |   |
|---|---|---|---|---|
| s | s | s | s | s |
| p | p | p | p |   |
| d | d | d |   |   |
| f | f |   |   |   |
| g |   |   |   |   |

aug-cc-pVQZ: 6s5p4d3f2g

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| s | s | s | s | s | s |
| p | p | p | p | p |   |
| d | d | d | d |   |   |
| f | f | f |   |   |   |
| g | g |   |   |   |   |

d-aug-cc-pVQZ: 7s6p5d4f3g

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| s | s | s | s | s | s | s |
| p | p | p | p | p | p |   |
| d | d | d | d | d |   |   |
| f | f | f | f |   |   |   |
| g | g | g |   |   |   |   |

# Core-valence correlation consistent basis for 2nd-row atoms

cc-pVDZ:  $3s2p1d = 1s + 2s2p1d$

|   |   |   |
|---|---|---|
| s | s | s |
| p | p |   |
| d |   |   |

cc-pCVDZ:  $4s3p1d = 2s1p + 2s2p1d$

|   |   |   |   |
|---|---|---|---|
| s | s | s | s |
| p | p | p |   |
| d |   |   |   |

cc-pVTZ:  $4s3p2d1f = 1s + 3s3p2s1f$

|   |   |   |   |
|---|---|---|---|
| s | s | s | s |
| p | p | p |   |
| d | d |   |   |
| f |   |   |   |

cc-pCVTZ:  $6s5p3d1f = 3s2p1d + 3s3p2d1f$

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| s | s | s | s | s | s |
| p | p | p | p | p |   |
| d | d | d |   |   |   |
| f |   |   |   |   |   |

cc-pVQZ:  $5s4p3d2f1g = 1s + 4s4p3d2f1g$

|   |   |   |   |   |
|---|---|---|---|---|
| s | s | s | s | s |
| p | p | p | p |   |
| d | d | d |   |   |
| f | f |   |   |   |
| g |   |   |   |   |

cc-pCVQZ:  $8s7p5d3f1g = 4s3p2d1f + 4s4p3d2f1g$

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| s | s | s | s | s | s | s | s |
| p | p | p | p | p | p | p |   |
| d | d | d | d | d |   |   |   |
| f | f | f |   |   |   |   |   |
| g |   |   |   |   |   |   |   |

# Jensen basis sets

pc-0: 3s2p (unpolarized)

s s s  
p p

pc-2: 4s3p2d1f

s s s s  
p p p  
d d  
f

pc-1: 3s2p1d (one set of polarization)

s s s  
p p  
d

pc-3: 6s5p4d2f1g

s s s s s s  
p p p p p  
d d d d  
f f  
g

## Polarization-consistent (pc) basis sets

- Designed for efficient convergence of HF and KS-DFT calculations
- Higher angular momenta are selected according to their energetic importance
- Diffuse variants: aug-pc-*n*

- Converge monotonically to the HF limit:

$$E_{\text{HF}}(L) = E_{\text{HF}}^{\text{CBS}} + A(L+1) \exp(-B\sqrt{L})$$

**Key idea:** Molecules are assemblies of slightly distorted atoms

## Advantages

- Compact representation of molecular orbitals
- Efficient analytic evaluation of multicenter integrals
- Practical all-electron calculations, especially for light and medium elements
- Locality enables integral screening and sparse algorithms
- Well suited to mean-field and correlated methods

## Limitations

- Nonorthogonal basis: possible linear-dependence problems
- Diffuse functions required for anions, Rydberg states, and long-range density tails
- Individual Gaussians have neither the correct nuclear cusp nor exponential asymptotic decay
- Atom-centered basis produces Pulay terms when nuclei move

⇒ Particularly effective for localized electronic structure correlated molecular calculations

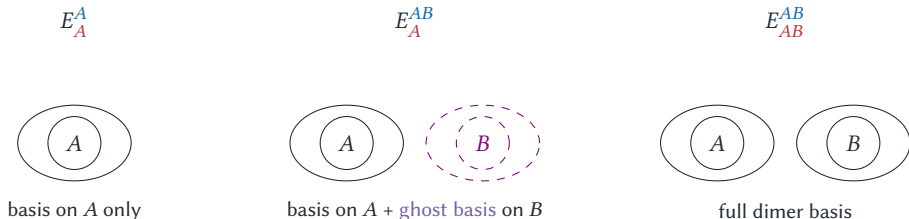
# Basis-set superposition error (BSSE)

## Notation

We denote by

$$E_X^Y = \text{energy of system } X \text{ computed in the basis of system } Y$$

For example,  $E_A^{AB}$  is the energy of fragment A computed using the basis functions centered on both A and B. The basis functions on B are then **ghost functions** (no nucleus and no electrons on B).



$$E_A^{AB} \leq E_A^A \Rightarrow \text{fragment A is artificially stabilized by the basis on B}$$

$\Rightarrow$  Fragments can “borrow” basis functions from each other

# Counterpoise correction

## Boys-Bernardi counterpoise correction

With a finite atom-centered basis, each fragment can **borrow basis functions from the other fragment**

$$E_A^{AB} \leq E_A^A \quad \text{and} \quad E_B^{AB} \leq E_B^B$$

The usual interaction energy is

$$\Delta E_{\text{int}} = E_{AB}^{AB} - E_A^A - E_B^B$$

while the counterpoise-corrected interaction energy is

$$\Delta E_{\text{int}}^{\text{CP}} = E_{AB}^{AB} - E_A^{AB} - E_B^{AB}$$

## Basis-set superposition error

$$\text{BSSE} = \Delta E_{\text{int}}^{\text{CP}} - \Delta E_{\text{int}} = \left( E_A^A - E_A^{AB} \right) + \left( E_B^B - E_B^{AB} \right) \geq 0$$

⇒ BSSE makes the uncorrected interaction energy artificially too attractive

# Effective core potentials (ECPs) or pseudopotentials

**Key idea:** Treat only the chemically relevant electrons explicitly

## One-electron ECP

Replace the core electrons by an **effective one-electron potential** acting on the remaining electrons

$$\hat{h}(\mathbf{r}) = -\frac{\nabla^2}{2} - \sum_A \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} \quad \longrightarrow \quad \boxed{\hat{h}^{\text{ECP}}(\mathbf{r}) = -\frac{\nabla^2}{2} - \sum_A \frac{Q_A}{|\mathbf{r} - \mathbf{R}_A|} + \sum_A \hat{U}_A^{\text{ECP}}(\mathbf{r})}$$

Here,  $Q_A = Z_A - N_A^{\text{core}}$  is an **effective nuclear charge** and  $N_A^{\text{core}}$  is the number of electrons replaced on atom  $A$

## Why use ECPs?

- ✓ **Lower cost:** fewer electrons and no basis functions needed to represent core orbitals
- ✓ **Relativistic effects:** can incorporate scalar-relativistic effects and, with a suitable operator, spin-orbit coupling
- ✓ Used in both molecular and periodic calculations, with Gaussian or plane-wave basis sets
- ✗ Introduce new **one-electron integrals** that can be tricky to compute
- ✗ Core properties are not described explicitly, and accuracy depends on the ECP quality



# ECPs: projected and unprojected terms

## A common semilocal form: scalar ECP

$$\hat{U}^{\text{ECP}}(\mathbf{r}) = U_L(\mathbf{r}) + \sum_{\ell=0}^{L-1} \sum_{m=-\ell}^{\ell} |Y_{\ell m}\rangle U_{\ell}(\mathbf{r}) \langle Y_{\ell m}|$$

- $U_L(\mathbf{r})$ : short-range **local** potential  $\Leftrightarrow$  **unprojected part**
- $\{|Y_{\ell m}\rangle U_{\ell}(\mathbf{r}) \langle Y_{\ell m}|\}_{0 \leq \ell < L}$ : angular-momentum-dependent **semi-local** potentials  $\Leftrightarrow$  **projected part**

## Gaussian expansion

$$U_{\ell}(\mathbf{r}) = \sum_{k=1}^{K_{\ell}} D_{\ell k} r^{n_{\ell k}} \exp(-\eta_{\ell k} r^2) \quad \ell = 0, \dots, L$$

- $D_{\ell k}$  and  $\eta_{\ell k} > 0$  are contraction coefficients and Gaussian exponents
- $n_{\ell k}$ : radial powers, often  $-2$ ,  $-1$ , or  $0$
- The angular projectors complicate the integrals, but they remain **one-electron integrals**

## Choosing an ECP: core size and families

Small core vs large core: Ag ( $Z = 47$ )

| Model      | Core electrons | Explicit electrons | Explicit shells          |
|------------|----------------|--------------------|--------------------------|
| Large core | 36             | 11                 | $4d^{10} 5s^1$           |
| Small core | 28             | 19                 | $4s^2 4p^6 4d^{10} 5s^1$ |

A **small-core ECP** retains semicore shells explicitly: more flexibility, at a higher computational cost

### Common ECP families used with Gaussian basis sets

- **LANL and SBK/SBKJC**: established families; LANL2DZ combines an ECP with a double- $\zeta$  valence basis
- **Stuttgart/Cologne**: energy-consistent pseudopotentials with different core sizes and dedicated basis sets <https://www.tc.uni-koeln.de/PP/clickpse.en.html>
- **def2-ECP**: paired with the def2 basis sets of Weigend and Ahlrichs
- **Burkatzki-Filippi-Dolg (BFD)**: developed for quantum Monte Carlo calculations
- **Correlation-consistent ECPs (ccECP)**: developed for high accuracy correlated many-body methods <https://pseudopotentiallibrary.org>

Most of them have dedicated basis sets!

# Properties of Gaussian functions

Gaussian product rule: “The product of two gaussians is a gaussian”

$$G_{\alpha, \mathbf{A}}(\mathbf{r}) = \exp\left(-\alpha|\mathbf{r} - \mathbf{A}|^2\right) \quad \text{and} \quad G_{\beta, \mathbf{B}}(\mathbf{r}) = \exp\left(-\beta|\mathbf{r} - \mathbf{B}|^2\right) \quad \text{then}$$

$$\boxed{G_{\alpha, \mathbf{A}}(\mathbf{r})G_{\beta, \mathbf{B}}(\mathbf{r}) = K_{\zeta, \mathbf{P}}G_{\zeta, \mathbf{P}}(\mathbf{r})} \quad \text{with} \quad \zeta = \alpha + \beta \quad \text{and} \quad \mathbf{P} = \frac{\alpha\mathbf{A} + \beta\mathbf{B}}{\alpha + \beta}$$

$$K_{\zeta, \mathbf{P}} = \exp\left(-\frac{\alpha\beta}{\alpha + \beta}|\mathbf{A} - \mathbf{B}|^2\right) \Rightarrow \text{decay exponentially fast with } |\mathbf{A} - \mathbf{B}|$$

## Overlap

$$\begin{aligned} \int G_{\alpha, \mathbf{A}}(\mathbf{r})G_{\beta, \mathbf{B}}(\mathbf{r}) \, d\mathbf{r} &= K_{\zeta, \mathbf{P}} \int G_{\zeta, \mathbf{P}}(\mathbf{r}) \, d\mathbf{r} \\ &= K_{\zeta, \mathbf{P}} \left( \int_{-\infty}^{+\infty} e^{-\zeta x^2} \, dx \right) \left( \int_{-\infty}^{+\infty} e^{-\zeta y^2} \, dy \right) \left( \int_{-\infty}^{+\infty} e^{-\zeta z^2} \, dz \right) \\ &= K_{\zeta, \mathbf{P}} \left( \frac{\pi}{\zeta} \right)^{3/2} \Rightarrow \text{cartesian factorization!} \end{aligned}$$

# One-electron integrals over Gaussian functions

## Kinetic energy

$$\begin{aligned}-\frac{1}{2}\nabla^2 G_{\beta,\mathbf{B}}(\mathbf{r}) &= [3\beta - 2\beta^2|\mathbf{r} - \mathbf{B}|^2]G_{\beta,\mathbf{B}}(\mathbf{r}) \\ -\frac{1}{2}\int G_{\alpha,\mathbf{A}}(\mathbf{r})\nabla^2 G_{\beta,\mathbf{B}}(\mathbf{r})\,d\mathbf{r} &= \int G_{\alpha,\mathbf{A}}(\mathbf{r})[3\beta - 2\beta^2|\mathbf{r} - \mathbf{B}|^2]G_{\beta,\mathbf{B}}(\mathbf{r})\,d\mathbf{r} \\ &= \frac{\alpha\beta}{\zeta}\left[3 - 2\frac{\alpha\beta}{\zeta}|\mathbf{A} - \mathbf{B}|^2\right]K_{\zeta,\mathbf{P}}\left(\frac{\pi}{\zeta}\right)^{3/2}\end{aligned}$$

## Nuclear attraction

For a nucleus of charge  $Z_C$  located at  $\mathbf{C}$

$$\begin{aligned}-Z_C\int G_{\alpha,\mathbf{A}}(\mathbf{r})\frac{1}{|\mathbf{r} - \mathbf{C}|}G_{\beta,\mathbf{B}}(\mathbf{r})\,d\mathbf{r} &= -Z_C K_{\zeta,\mathbf{P}}\int\frac{e^{-\zeta|\mathbf{r} - \mathbf{P}|^2}}{|\mathbf{r} - \mathbf{C}|}\,d\mathbf{r} \\ &= -\frac{2Z_C}{\sqrt{\pi}}K_{\zeta,\mathbf{P}}\int_0^\infty dt\int e^{-\zeta|\mathbf{r} - \mathbf{P}|^2}e^{-t^2|\mathbf{r} - \mathbf{C}|^2}\,d\mathbf{r} \\ &= -\frac{2\pi Z_C}{\zeta}K_{\zeta,\mathbf{P}}F_0\left(\zeta|\mathbf{P} - \mathbf{C}|^2\right)\end{aligned}$$

# Four-center electron repulsion integrals

## Gaussian product rule for ERIs

Formally, one has to compute  $O(K^4)$  ERIs!

$$\begin{aligned}(\mathbf{ab}|\mathbf{cd}) &= \iint G_{\alpha,\mathbf{A}}(\mathbf{r}_1)G_{\beta,\mathbf{B}}(\mathbf{r}_1)\frac{1}{r_{12}}G_{\gamma,\mathbf{C}}(\mathbf{r}_2)G_{\delta,\mathbf{D}}(\mathbf{r}_2)\mathrm{d}\mathbf{r}_1\mathrm{d}\mathbf{r}_2 \\ &= K_{\zeta,\mathbf{P}}K_{\eta,\mathbf{Q}}\iint G_{\zeta,\mathbf{P}}(\mathbf{r}_1)\frac{1}{r_{12}}G_{\eta,\mathbf{Q}}(\mathbf{r}_2)\mathrm{d}\mathbf{r}_1\mathrm{d}\mathbf{r}_2\end{aligned}$$

The number of “significant” ERIs in a large system is  $O(K^2)$ !

## Shell pairs and shell quartets

- $(\mathbf{ab}|$  = single primitive bra and  $(\mathbf{ab}|$  = entire primitive shell pair
- $(\mathbf{ab}|\mathbf{cd})$  = single primitive integral and  $(\mathbf{ab}|\mathbf{cd})$  = entire primitive shell quartet
- $(\mathbf{ab}|$  = single contracted bra and  $(\mathbf{ab}|$  = entire contracted shell pair
- $(\mathbf{ab}|\mathbf{cd})$  = single contracted integral and  $(\mathbf{ab}|\mathbf{cd})$  = entire contracted shell quartet

# Fundamental ssss integral

## Definition

$$\begin{aligned} [ss|ss] &= K_{\zeta, \mathbf{P}} K_{\eta, \mathbf{Q}} \iint G_{\zeta, \mathbf{P}}(\mathbf{r}_1) \frac{1}{r_{12}} G_{\eta, \mathbf{Q}}(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2 \\ &= K_{\zeta, \mathbf{P}} K_{\eta, \mathbf{Q}} \iint e^{-\zeta|\mathbf{r}_1 - \mathbf{P}|^2} \frac{1}{r_{12}} e^{-\eta|\mathbf{r}_2 - \mathbf{Q}|^2} d\mathbf{r}_1 d\mathbf{r}_2 \end{aligned}$$

## Laplace transform of the Coulomb operator

$$\frac{1}{r_{12}} = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2 r_{12}^2} dt = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2(x_1-x_2)^2} e^{-t^2(y_1-y_2)^2} e^{-t^2(z_1-z_2)^2} dt$$

$$\begin{aligned} [ss|ss] &= \frac{2}{\sqrt{\pi}} K_{\zeta, \mathbf{P}} K_{\eta, \mathbf{Q}} \int_0^\infty dt \left( \int_{-\infty}^\infty \int_{-\infty}^\infty e^{-\zeta(x_1 - \mathbf{P}_x)^2} e^{-t^2(x_1-x_2)^2} e^{-\eta(x_2 - \mathbf{Q}_x)^2} dx_1 dx_2 \right) \\ &\quad \times \left( \int_{-\infty}^\infty \int_{-\infty}^\infty e^{-\zeta(y_1 - \mathbf{P}_y)^2} e^{-t^2(y_1-y_2)^2} e^{-\eta(y_2 - \mathbf{Q}_y)^2} dy_1 dy_2 \right) \\ &\quad \times \left( \int_{-\infty}^\infty \int_{-\infty}^\infty e^{-\zeta(z_1 - \mathbf{P}_z)^2} e^{-t^2(z_1-z_2)^2} e^{-\eta(z_2 - \mathbf{Q}_z)^2} dz_1 dz_2 \right) \end{aligned}$$

# Fundamental ssss integral

Defining  $\vartheta = \zeta\eta/(\zeta + \eta)$ , we have

$$\begin{aligned}
 [ss|ss] &= \frac{2}{\sqrt{\pi}} K_{\zeta, \mathbf{P}} K_{\eta, \mathbf{Q}} \int_0^\infty \left( \frac{\pi}{\sqrt{\zeta\eta}} \frac{e^{-\vartheta(\mathbf{P}_x - \mathbf{Q}_x)^2 \frac{t^2}{t^2 + \vartheta}}}{\sqrt{1 + \frac{t^2}{\vartheta}}} \right) \left( \frac{\pi}{\sqrt{\zeta\eta}} \frac{e^{-\vartheta(\mathbf{P}_y - \mathbf{Q}_y)^2 \frac{t^2}{t^2 + \vartheta}}}{\sqrt{1 + \frac{t^2}{\vartheta}}} \right) \left( \frac{\pi}{\sqrt{\zeta\eta}} \frac{e^{-\vartheta(\mathbf{P}_z - \mathbf{Q}_z)^2 \frac{t^2}{t^2 + \vartheta}}}{\sqrt{1 + \frac{t^2}{\vartheta}}} \right) dt \\
 &= \frac{2\pi^{5/2}}{(\zeta\eta)^{3/2}} K_{\zeta, \mathbf{P}} K_{\eta, \mathbf{Q}} \int_0^\infty \frac{e^{-\vartheta|\mathbf{P} - \mathbf{Q}|^2 \frac{t^2}{t^2 + \vartheta}}}{\left(1 + \frac{t^2}{\vartheta}\right)^{3/2}} dt
 \end{aligned}$$

$$\text{Setting } t^2 = \vartheta \frac{u^2}{1 - u^2} \quad \Rightarrow \quad [ss|ss] = \sqrt{\frac{4\vartheta}{\pi}} \left(\frac{\pi}{\zeta}\right)^{3/2} K_{\zeta, \mathbf{P}} \left(\frac{\pi}{\eta}\right)^{3/2} K_{\eta, \mathbf{Q}} \underbrace{\int_0^1 e^{-\vartheta|\mathbf{P} - \mathbf{Q}|^2 u^2} du}_{F_0(\vartheta|\mathbf{P} - \mathbf{Q}|^2) = \text{Boys function}}$$

$$[00|00]^{(0)} = \sqrt{\frac{4\vartheta}{\pi}} \left(\frac{\pi}{\zeta}\right)^{3/2} K_{\zeta, \mathbf{P}} \left(\frac{\pi}{\eta}\right)^{3/2} K_{\eta, \mathbf{Q}} F_0(\vartheta|\mathbf{P} - \mathbf{Q}|^2)$$

# Boys function

## Generalized Boys function

$$F_m(x) = \int_0^1 t^{2m} \exp(-xt^2) dt \quad F_0(x) = \sqrt{\frac{\pi}{4x}} \operatorname{erf}(\sqrt{x})$$

## Computation

- $F_m(x)$  computed using pretabulated values on a grid and Taylor expansion (Chebyshev interpolation table)
- higher  $m$  values computed via downward recursion

## Downward recursion

$$F_m(x) = \frac{2xF_{m+1}(x) + \exp(-x)}{2m+1}$$

Downward recursion is numerically stable, whereas upward recursion can become unstable for small  $x$

## Asymptotic behavior

$$F_m(x) \sim \frac{(2m-1)!!}{2^{m+1}} \sqrt{\frac{\pi}{x^{2m+1}}} \quad \text{for large } x$$



### Why do higher-order Boys functions appear?

The physical ssss ERI involves only  $F_0$ .

However, generating angular momentum by differentiation produces higher-order Boys functions  $F_m$ .

It is therefore convenient to define the hierarchy

$$[00|00]^{(m)} = \sqrt{\frac{4\vartheta}{\pi}} \left(\frac{\pi}{\zeta}\right)^{3/2} K_{\zeta, \mathbf{P}} \left(\frac{\pi}{\eta}\right)^{3/2} K_{\eta, \mathbf{Q}} F_m(\vartheta |\mathbf{P} - \mathbf{Q}|^2)$$

- $m = 0$ : physical ssss ERI
- $m > 0$ : auxiliary integrals, used only as intermediates
- Angular-momentum recurrences couple auxiliary orders  $m$  and  $m + 1$
- For total angular momentum  $L = a + b + c + d$   
 $\Rightarrow$  one needs fundamental auxiliary integrals up to  $m = L$ .

### Key identity

For a Cartesian Gaussian,

$$\frac{\partial |\mathbf{a}|}{\partial A_i} = 2\alpha |\mathbf{a} + \mathbf{1}_i| - a_i |\mathbf{a} - \mathbf{1}_i| \quad \Rightarrow \quad |\mathbf{a} + \mathbf{1}_i| = \frac{1}{2\alpha} \left[ \frac{\partial |\mathbf{a}|}{\partial A_i} + a_i |\mathbf{a} - \mathbf{1}_i| \right]$$

Applying this identity to the auxiliary ERIs generates recurrence relations connecting different angular momenta and different Boys orders  $\Rightarrow$  **Obara-Saika or vertical recurrence relations**

$$[00|00]^{(m)} \longrightarrow [a0|00]^{(m)} \longrightarrow [a0|c0]^{(m)}$$

Instead of evaluating every ERI from scratch, we build them recursively

# Strategy for building an ERI

Starting from the fundamental auxiliary integrals,

$$\boxed{[00|00]^{(m)} \xrightarrow{\text{VRR}} [a0|c0]^{(m)} \xrightarrow{\text{contraction}} (a0|c0) \xrightarrow{\text{HRR}} (ab|cd)}$$

- **Vertical recurrence relations (VRRs)**: create angular momentum on one center of each Gaussian pair
- **Contraction**: combine primitive integrals into contracted integrals
- **Horizontal recurrence relations (HRRs)**: redistribute angular momentum within each Gaussian pair

VRRs create angular momentum

HRRs redistribute it

VRR on the bra pair: build angular momentum on  $A$

$$\begin{aligned}
 [(\mathbf{a} + \mathbf{1}_i)\mathbf{0}|\mathbf{00}]^{(m)} &= (P_i - A_i)[\mathbf{a}\mathbf{0}|\mathbf{00}]^{(m)} \\
 &\quad + \frac{\eta}{\zeta + \eta} (P_i - Q_i)[\mathbf{a}\mathbf{0}|\mathbf{00}]^{(m+1)} \\
 &\quad + \frac{a_i}{2\zeta} \left[ [(\mathbf{a} - \mathbf{1}_i)\mathbf{0}|\mathbf{00}]^{(m)} - \frac{\eta}{\zeta + \eta} [(\mathbf{a} - \mathbf{1}_i)\mathbf{0}|\mathbf{00}]^{(m+1)} \right]
 \end{aligned}$$

VRR on the ket pair: build angular momentum on  $C$

$$\begin{aligned}
 [\mathbf{a}\mathbf{0} | (\mathbf{c} + \mathbf{1}_i)\mathbf{0}]^{(m)} &= (Q_i - C_i)[\mathbf{a}\mathbf{0} | \mathbf{c}\mathbf{0}]^{(m)} \\
 &\quad + \frac{\zeta}{\zeta + \eta} (Q_i - P_i)[\mathbf{a}\mathbf{0} | \mathbf{c}\mathbf{0}]^{(m+1)} \\
 &\quad + \frac{c_i}{2\eta} \left[ [\mathbf{a}\mathbf{0} | (\mathbf{c} - \mathbf{1}_i)\mathbf{0}]^{(m)} - \frac{\zeta}{\zeta + \eta} [\mathbf{a}\mathbf{0} | (\mathbf{c} - \mathbf{1}_i)\mathbf{0}]^{(m+1)} \right] \\
 &\quad + \frac{a_i}{2(\zeta + \eta)} [(\mathbf{a} - \mathbf{1}_i)\mathbf{0} | \mathbf{c}\mathbf{0}]^{(m+1)}
 \end{aligned}$$

## Horizontal recurrence relations

### Redistributing angular momentum

Once angular momentum has been generated on one center of a Gaussian pair, it can be shifted to the other center:

$$[a(b + 1_i)|cd] = [(a + 1_i)b|cd] + (A_i - B_i)[ab|cd]$$

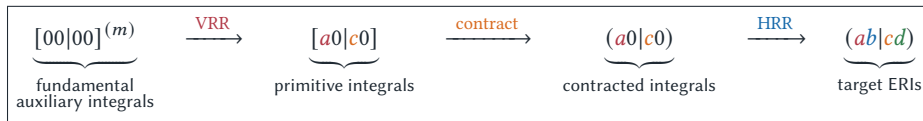
and, similarly, angular momentum can be transferred from  $C$  to  $D$

### Why apply HRRs after contraction?

The HRR depends only on the geometry through  $A_i - B_i$ : it is **independent of the Gaussian exponents**  
Therefore, the same recurrence applies directly to contracted integrals

$$(a(b + 1_i)|cd) = ((a + 1_i)b|cd) + (A_i - B_i)(ab|cd)$$

# Late-contraction ERI algorithm



1. Compute the required Boys functions  $F_m(T)$
2. Form the fundamental auxiliary integrals  $[00|00]^{(m)}$
3. Use **VRRs** to build angular momentum on  $A$  and  $C$
4. **Contract** the primitive integrals
5. Use exponent-independent **HRRs** to transfer angular momentum  $A \rightarrow B$  and  $C \rightarrow D$

Expensive exponent-dependent recurrences are performed before contraction, while the cheap exponent-independent HRRs are performed afterwards

# Upper bounds for ERIs

A “good” upper bound must be

- simple, i.e, cheap to compute
- strong, i.e., a good estimate
- scaling-consistent, i.e.,  $N_{\text{sig}} = O(N_{\text{upper}})$

Cauchy-Schwartz bound

$$|(ab|cd)| \leq \sqrt{(ab|ab)}\sqrt{(cd|cd)} \quad \text{or} \quad |(P|Q)| \leq \sqrt{(P|P)}\sqrt{(Q|Q)}$$

- $(P|Q)$  should decay as  $1/|P - Q|$  or faster
- However, there is no distance information in the product  $(P|P)(Q|Q)$
- The Cauchy-Schwarz bound is rigorous and cheap but is not tight for well-separated shell pairs

⇒ A cheap upper bound avoids the evaluation of negligible ERIs

## Integral bound

We seek a single number that bounds a particular integral

$$|(ab|cd)| \leq \sqrt{(ab|ab)}\sqrt{(cd|cd)}$$

With  $O(K^2)$   $(ab|ab)$  bound factors, we can upper bound  $O(K^4)$   $(ab|cd)$  integrals

## Class bound

We seek a single number that bounds an entire class

$$|(ab|cd)| \leq Q_{ab}Q_{cd} \quad \text{with} \quad Q_{ab} = \max_{\substack{a \in a \\ b \in b}} \sqrt{(ab|ab)}$$

Therefore,

if  $Q_{ab}Q_{cd} < \tau$  then **discard the entire shell quartet!**

\*The simple  $(dd|dd)$  class is made of 1296 integrals!



# Asymptotic scaling of two-electron integrals

## Number of significant two-electron integrals

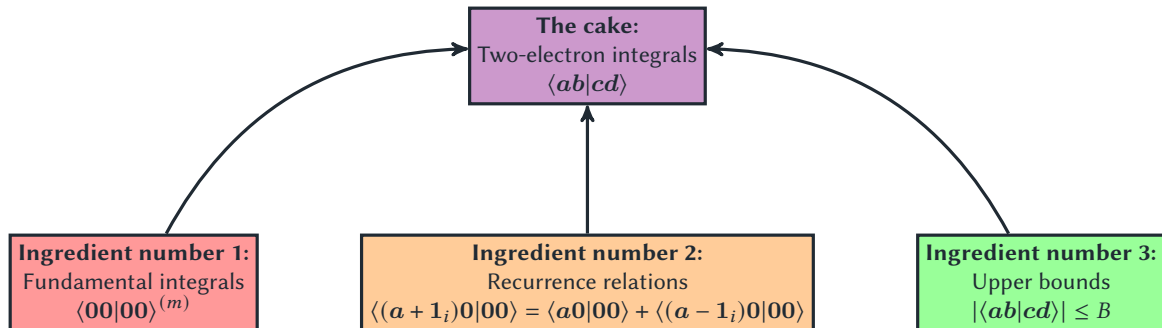
$$(ab|cd) \equiv (ab|\hat{O}_2|cd)$$

## Long-range vs short-range operators

$$N_{\text{sig}} = c N^{\alpha}$$

| Molecule     | N  | $\hat{O} = r_{12}^{-1}$ |          | $\hat{O} = e^{-r_{12}^2}$ |          |
|--------------|----|-------------------------|----------|---------------------------|----------|
|              |    | $N_{\text{sig}}$        | $\alpha$ | $N_{\text{sig}}$          | $\alpha$ |
| propene      | 12 | 1 625                   | —        | 1 650                     | —        |
| butadiene    | 16 | 5 020                   | 3.9      | 5 020                     | 3.9      |
| hexatriene   | 24 | 24 034                  | 3.9      | 23 670                    | 3.8      |
| octatetraene | 32 | 63 818                  | 3.4      | 52 808                    | 2.8      |
| decapentaene | 40 | 119 948                 | 2.8      | 81 404                    | 1.9      |
| dodecaexaene | 48 | 192 059                 | 2.6      | 109 965                   | 1.6      |

# Recipe for computing two-electron integrals



# Direct SCF algorithm

## From ERIs to the Fock matrix

The Coulomb and exchange matrices are

$$J_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\nu|\lambda\sigma)$$

$$K_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\lambda|\nu\sigma)$$

## Direct SCF: trading disk storage for repeated integral evaluation

Instead of storing all  $O(K^4)$  ERIs

1. loop over shell quartets  $(\mu\nu|\lambda\sigma)$
2. use an upper bound to discard negligible quartets
3. compute the significant ERIs
4. immediately accumulate their contributions to  $\mathbf{J}$  and  $\mathbf{K}$
5. discard the ERIs

The density matrix elements can be included in the screening process

screen  $\longrightarrow$  compute  $\longrightarrow$  accumulate  $\longrightarrow$  discard

# Resolution of the identity (RI)

## The product space is the problem

An ERI is the Coulomb interaction between two **pair densities**,

$$(\mu\nu|\lambda\sigma) = \iint \underbrace{\phi_\mu(\mathbf{r}_1)\phi_\nu(\mathbf{r}_1)}_{\rho_{\mu\nu}(\mathbf{r}_1)} \frac{1}{r_{12}} \underbrace{\phi_\lambda(\mathbf{r}_2)\phi_\sigma(\mathbf{r}_2)}_{\rho_{\lambda\sigma}(\mathbf{r}_2)} d\mathbf{r}_1 d\mathbf{r}_2$$

With  $K$  basis functions, there are formally  $O(K^2)$  different pair densities  $\rho_{\mu\nu}$

## Density fitting

Instead of representing each product  $\phi_\mu\phi_\nu$  exactly, we expand it in a smaller **auxiliary basis**  $\{\varphi_A^{\text{aux}}\}$ ,

$$\rho_{\mu\nu}(\mathbf{r}) \approx \tilde{\rho}_{\mu\nu}(\mathbf{r}) = \sum_A C_{\mu\nu}^A \varphi_A^{\text{aux}}(\mathbf{r})$$

with typically  $K_{\text{aux}} = O(K)$

$\Rightarrow$  RI replaces the large AO-product space by a compact auxiliary space

### How should the pair density be fitted?

Define the Coulomb inner product

$$(f|g) = \iint f(\mathbf{r}_1) \frac{1}{r_{12}} g(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2$$

The optimal coefficients minimize the Coulomb norm of the fitting error,

$$\Delta_{\mu\nu} = (\rho_{\mu\nu} - \tilde{\rho}_{\mu\nu} | \rho_{\mu\nu} - \tilde{\rho}_{\mu\nu})$$

### Least-squares equations

Minimization with respect to  $C_{\mu\nu}^A$  gives

$$\sum_B \underbrace{(A|B)}_{V_{AB}} C_{\mu\nu}^B = \underbrace{(A|\mu\nu)}_{\text{three-center integral}} \Rightarrow \boxed{C_{\mu\nu}^A = \sum_B (\mathbf{V}^{-1})_{AB} (B|\mu\nu)}$$

$\Rightarrow$  The Coulomb metric determines the optimal auxiliary expansion

# RI approximation to the ERIs

## From four-center to three-center integrals

Inserting the fitted pair densities gives

$$(\mu\nu|\lambda\sigma) \approx \sum_{AB} (\mu\nu|A) \left( \mathbf{V}^{-1} \right)_{AB} (B|\lambda\sigma)$$

## Symmetric factorization

Defining the orthonormalized three-index quantities

$$B_{\mu\nu}^P = \sum_A (\mu\nu|A) \left( \mathbf{V}^{-1/2} \right)_{AP}$$

one obtains the particularly simple form

$$(\mu\nu|\lambda\sigma) \approx \sum_P B_{\mu\nu}^P B_{\lambda\sigma}^P$$

⇒ A four-index tensor is factorized into products of three-index tensors

## Low-rank representations of the ERIs

### Resolution of the identity = Density fitting

In RI/DF, the AO pair densities are expanded in a **predefined auxiliary basis**  $\{\chi_A^{\text{aux}}\}$ ,

$$\chi_\mu(\mathbf{r})\chi_\nu(\mathbf{r}) \approx \sum_A C_{\mu\nu}^A \chi_A^{\text{aux}}(\mathbf{r}) \quad \Rightarrow \quad (\mu\nu|\lambda\sigma) \approx \sum_P B_{\mu\nu}^P B_{\lambda\sigma}^P$$

$\Rightarrow$  The quality of the approximation depends on the auxiliary basis

### Cholesky decomposition (CD)

The ERI tensor  $V_{(\mu\nu),(\lambda\sigma)} = (\mu\nu|\lambda\sigma)$  can be viewed as a positive semidefinite matrix in the compound indices  $(\mu\nu)$  and  $(\lambda\sigma)$ . A pivoted Cholesky decomposition directly generates

$$(\mu\nu|\lambda\sigma) \approx \sum_{P=1}^{N_{\text{CD}}} L_{\mu\nu}^P L_{\lambda\sigma}^P$$

until the remaining error is smaller than a prescribed threshold  $\tau_{\text{CD}}$

$\Rightarrow$  RI/DF uses a predefined auxiliary space; CD constructs one adaptively

## Construction of the Coulomb matrix

Starting from

$$J_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\nu|\lambda\sigma),$$

the RI approximation gives

$$J_{\mu\nu} \approx \sum_{AB} (\mu\nu|A) \left( \mathbf{V}^{-1} \right)_{AB} \underbrace{\sum_{\lambda\sigma} P_{\lambda\sigma} (B|\lambda\sigma)}_{d_B}.$$

Thus,

$$\boxed{\mathbf{P} \longrightarrow \{d_A\} \longrightarrow \{c_A\} = \mathbf{V}^{-1}\{d_A\} \longrightarrow \mathbf{J}}$$

- No need to store the full  $\mathcal{O}(K^4)$  ERI tensor
- Only **two-center** ( $A|B$ ) and **three-center** ( $\mu\nu|A$ ) integrals are required
- Auxiliary basis sets are optimized to make the fitting error very small
- RI is also commonly called **density fitting (DF)**  
 $\Rightarrow$  RI dramatically reduces the cost and memory requirements of Coulomb contractions